Name:_____

Exam Question Booklets
OCR Physics A

Booklet 7
11 Questions
104 marks

Things to remember:

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1

In this question, two marks are available for the quality of written communication.

Fig. 2.1 shows the path of a ball after it is thrown from **T**. The ball reaches a maximum height at point **P** and then returns to the ground at **G**.

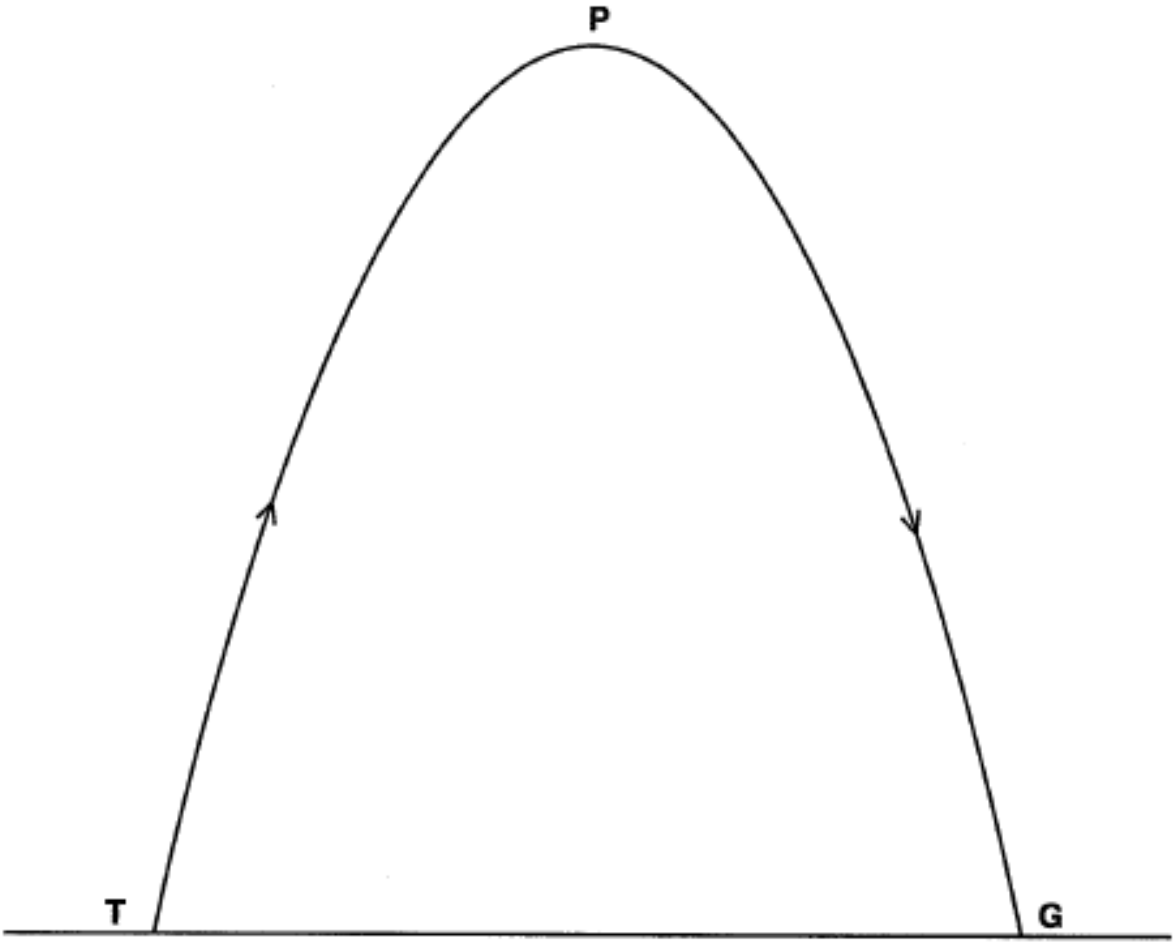

Fig. 2.1

(a) Assuming no air resistance, describe and explain how the vertical and horizontal components of the velocity of the ball change as it travels from **T** to **G**.

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..... [5]

(b) Assuming no air resistance, describe the changes in the kinetic and potential energies of the ball as it travels from **T** to **G**.

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.....[4]

(c) Describe how the motion of the ball is affected when air resistance is taken into consideration.

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.....[3]

Fig. 5.1 shows a lorry travelling up a slope.

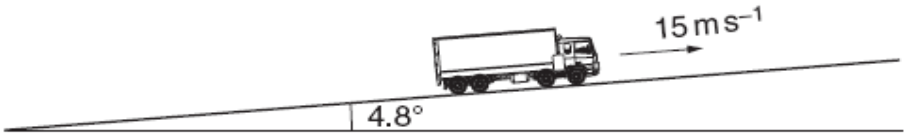

Fig. 5.1

The angle of the slope is 4.8° with the horizontal and the weight of the lorry is 2.4×10^5 N. A resistive force of 1.2×10^4 N down the slope acts on the lorry as it travels up the slope at a constant speed of $15\,\mathrm{m\,s^{-1}}$.

(a) Show that the component of the weight of the lorry down the slope is 2.0×10^4 N.

[1]

(b) Calculate the rate at which the lorry does work against the resistive force.

rate of work done = ... $\mathrm{J\,s^{-1}}$ [2]

(c) Calculate the power developed by the lorry as it travels up the slope.

power = ... W [2]

(d) Calculate the rate of gain of potential energy of the lorry.

rate of gain of potential energy = ... unit ..... [3]

(e) State and explain how the braking distance of the lorry up the slope compares with that on a horizontal road at the same speed.

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.....[1]

3 Fig. 5.1 shows part of the force-extension graph for a spring. The spring obeys Hooke's law for forces up to 5.0 N.

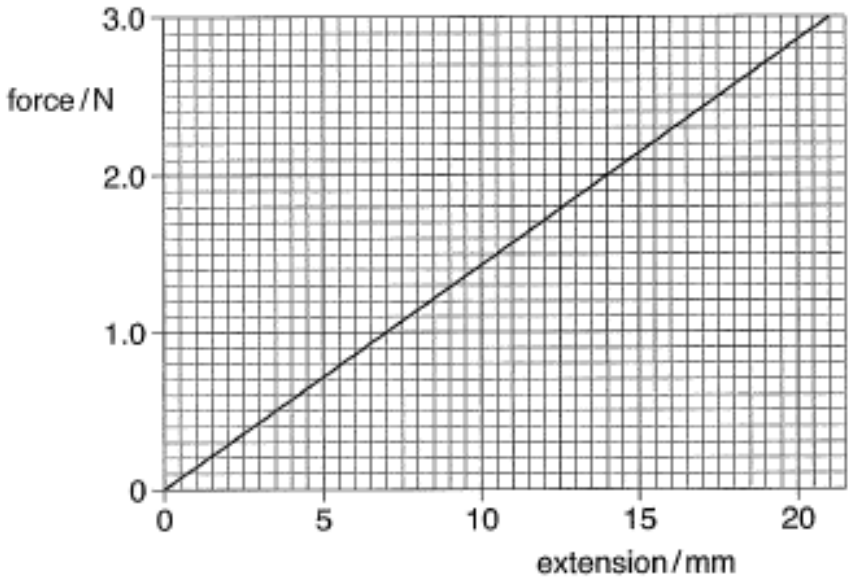

Fig. 5.1

(a) Calculate the extension produced by a force of 5.0 N.

extension = ...mm [2]

(b) Fig. 5.2 shows a second identical spring that has been put in parallel with the first spring A force of 5.0 N is applied to this combination of springs.

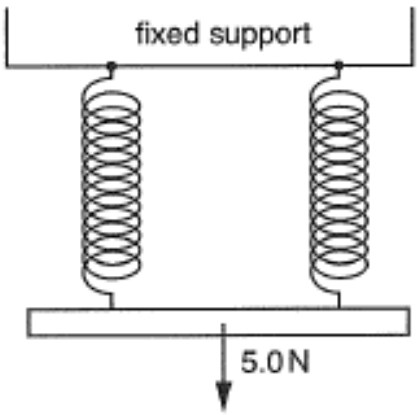

Fig. 5.2

For the arrangement shown in Fig. 5.2, calculate

(i) the extension of each spring

extension = ...mm [2]

(ii) the total strain energy stored in the springs.

strain energy = ...J [2]

(c) The Young modulus of the wire used in the springs is 2.0×10^{11} Pa. Each spring is made from a straight wire of length 0.40 m and cross-sectional area 2.0×10^{-7} m^2. Calculate the extension produced when a force of 5.0 N is applied to this straight wire.

extension = ...m [3]

(d) Describe and explain, without further calculations, the difference in the strain energies stored in the straight wire and in the spring when a 5.0 N force is applied to each.

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.....[2]

(a) **(i)** Define the *momentum* of a body.

.....[1]

(ii) A body, initially at rest, explodes into two unequal fragments of mass m_1 and m_2. Mass m_1 has a velocity v_1 and mass m_2 has a velocity v_2. Using the principle of conservation of momentum, derive an expression for v_1/v_2.

[2]

(b) An isolated nucleus of mass 4.0×10^{-25} kg is initially at rest. It decays, emitting an alpha particle of mass 6.7×10^{-27} kg with kinetic energy of 1.2×10^{-14} J.

(i) Show that the speed of the alpha particle is about 2×10^{8} m s^{-1}.

[2]

(ii) Calculate the momentum of the alpha particle.

momentum = ...kg m s^{-1} [1]

(iii) Hence find the speed of the recoiling nucleus.

speed = ...m s^{-1} [2]

(c) Before the decay described in **(b)** the nucleus is at point P on Fig. 1.1.

Fig. 1.1

(i) Place a small cross on Fig. 1.1 at a possible position, to full scale, of the alpha particle 8.0×10^{-9} s after the emission. [1]

(ii) Indicate with an arrow, starting at P, the direction of movement of the recoiling nucleus. [1]

(iii) Estimate how far the recoiling nucleus has moved in 8.0×10^{-9} s.

distance = ...mm [1]

(a) Fig. 1.1 shows a ball of mass 0.050 kg resting on the strings of a tennis racket held horizontally.

Fig. 1.1

(i) Draw and label the **two** forces acting on the ball. [2]

(ii) Each of these forces has a corresponding equal and opposite force according to Newton's third law of motion. Describe these equal and opposite forces and state the objects on which they act.

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..... [4]

(iii) Calculate the difference in magnitude between the two forces on the ball when the racket is accelerated upwards at $2.0\,m\,s^{-2}$.

force = ... N [2]

(b) The ball is dropped from rest at a point 0.80 m above the racket head. The racket is fixed rigidly. Assume that the ball makes an elastic collision with the strings and that any effects of air resistance are negligible.

Calculate

(i) the speed of the ball just before the impact

speed = ... $m\,s^{-1}$ [2]

(ii) the momentum of the ball just before impact

momentum = ... $kg\,m\,s^{-1}$ [1]

(iii) the change in momentum of the ball during the impact

momentum change = ... $kg\,m\,s^{-1}$ [1]

(iv) the average force during the impact for a contact time of 0.050 s.

force = ... N [1]

6

Two wires P and Q of the same material and same original length l_0 are fixed so that they hang vertically, as shown in Fig. 5.1.

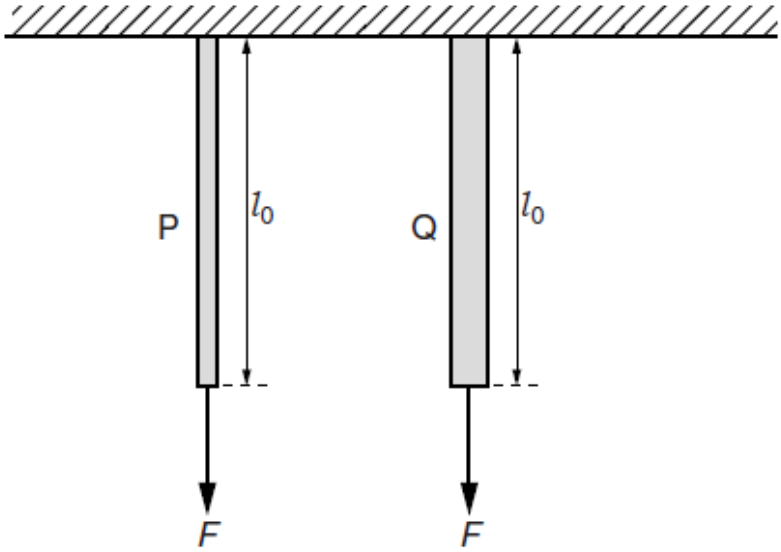

Fig. 5.1 (not to scale)

The diameter of P is d and the diameter of Q is $2d$. The same force F is applied to the lower end of each wire.

Show your working and determine the ratio

(i) $\dfrac{\text{stress in P}}{\text{stress in Q}}$,

ratio = ... [2]

(ii) $\dfrac{\text{strain in P}}{\text{strain in Q}}$.

ratio = ... [2]

7

A ball of mass 150 g is thrown vertically upwards with an initial speed of $25\,m\,s^{-1}$.

(i) Calculate the initial kinetic energy of the ball.

kinetic energy = ... J [3]

(ii) The ball reaches a height of 21 m above the point of release.

For the ball rising to this height, calculate

1. the loss of energy of the ball to air resistance,

energy loss = ... J [3]

2. the average force due to the air resistance.

force = ... N [2]

A ball is thrown vertically down towards the ground and rebounds as illustrated in Fig. 2.1.

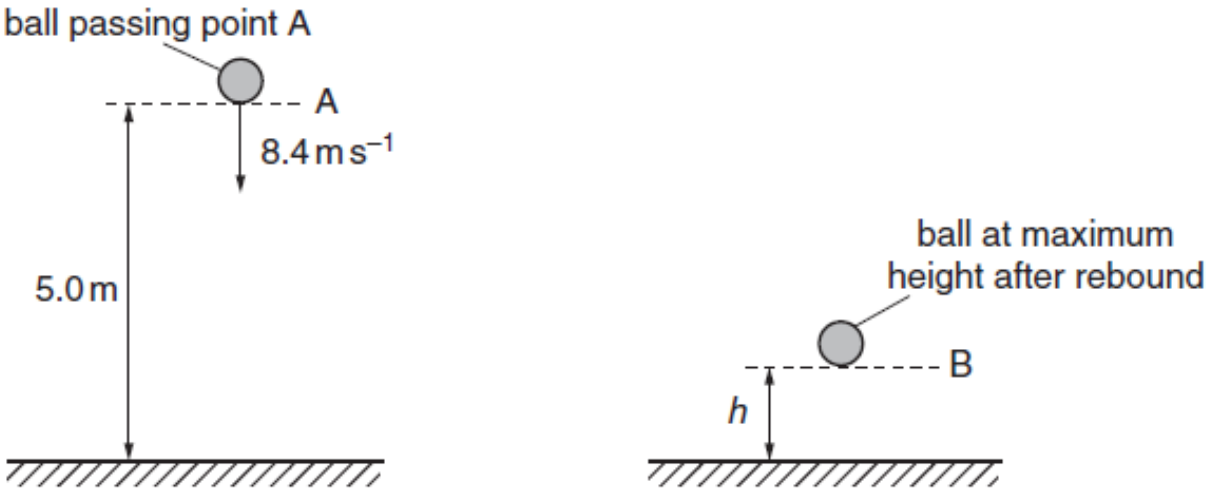

Fig. 2.1

As the ball passes A, it has a speed of $8.4\,m\,s^{-1}$. The height of A is 5.0 m above the ground. The ball hits the ground and rebounds to B. Assume that air resistance is negligible.

(i) Calculate the speed of the ball as it hits the ground.

speed = ... $m\,s^{-1}$ [2]

(ii) Show that the time taken for the ball to reach the ground is 0.47 s.

[1]

(b) The ball rebounds vertically with a speed of $4.2\,\mathrm{m\,s^{-1}}$ as it leaves the ground. The time the ball is in contact with the ground is 20 ms. The ball rebounds to a maximum height h.

The ball passes A at time $t = 0$. On Fig. 2.2, plot a graph to show the variation with time t of the velocity v of the ball. Continue the graph until the ball has rebounded from the ground and reaches B.

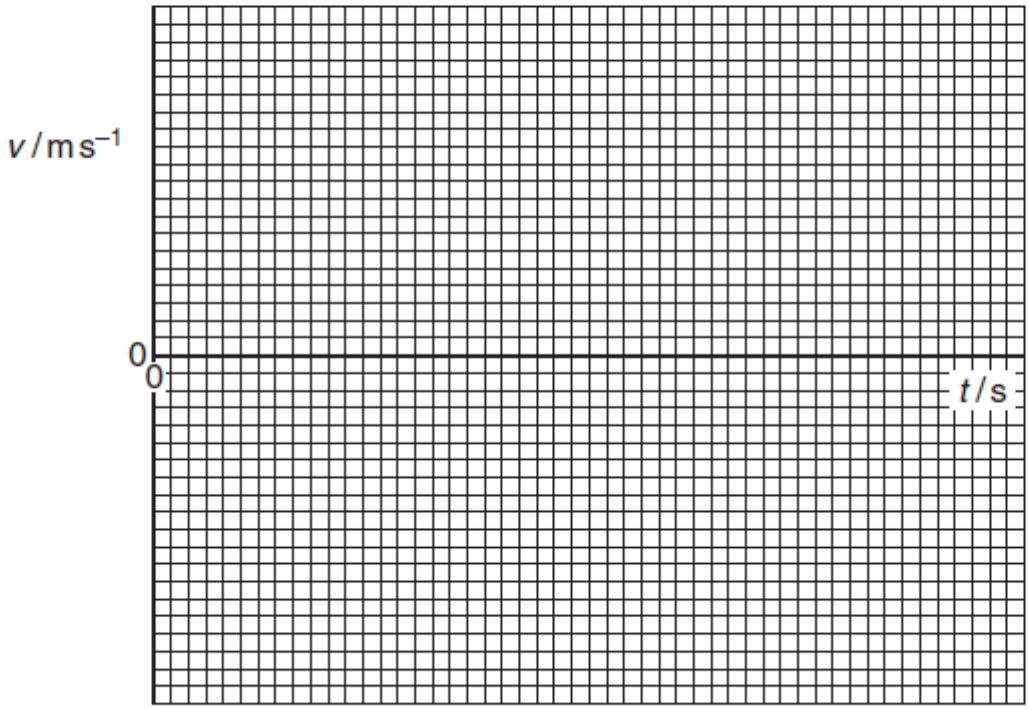

Fig. 2.2 [3]

(c) The ball has a mass of 0.050 kg. It moves from A and reaches B after rebounding.

(i) For this motion, calculate the change in

1. kinetic energy,

change in kinetic energy = ... J [2]

2. gravitational potential energy.

change in potential energy = ... J [3]

(ii) State and explain the total change in energy of the ball for this motion.

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.....[2]

A log of mass 450 kg is pulled up a slope by a wire attached to a motor, as shown in Fig. 3.1.

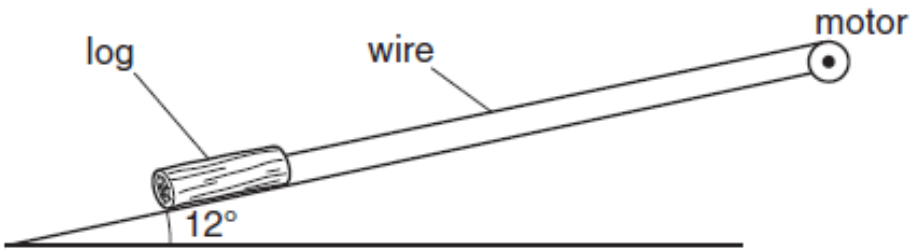

Fig. 3.1

The angle that the slope makes with the horizontal is 12°. The frictional force acting on the log is 650 N. The log travels with constant velocity.

(i) With reference to the motion of the log, discuss whether the log is in equilibrium.

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.....[2]

(ii) Calculate the tension in the wire.

tension = ... N [3]

(iii) State and explain whether the gain in the potential energy per unit time of the log is equal to the output power of the motor.

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.....[2]

A ball is thrown vertically down towards the ground with an initial velocity of $4.23\,m\,s^{-1}$. The ball falls for a time of 1.51 s before hitting the ground. Air resistance is negligible.

(a) **(i)** Show that the downwards velocity of the ball when it hits the ground is $19.0\,m\,s^{-1}$.

[2]

(ii) Calculate, to three significant figures, the distance the ball falls to the ground.

distance = ... m [2]

(b) The ball makes contact with the ground for 12.5 ms and rebounds with an upwards velocity of $18.6\,m\,s^{-1}$. The mass of the ball is 46.5 g.

(i) Calculate the average force acting on the ball on impact with the ground.

magnitude of force = ... N

direction of force .....

[4]

(ii) Use conservation of energy to determine the maximum height the ball reaches after it hits the ground.

height = ... m [2]

(c) State and explain whether the collision the ball makes with the ground is elastic or inelastic.

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.....[1]

A stone is thrown with a horizontal velocity of $20\,m\,s^{-1}$ from the top of a cliff 15 m high. The path of the stone is shown in Fig. 1.1.

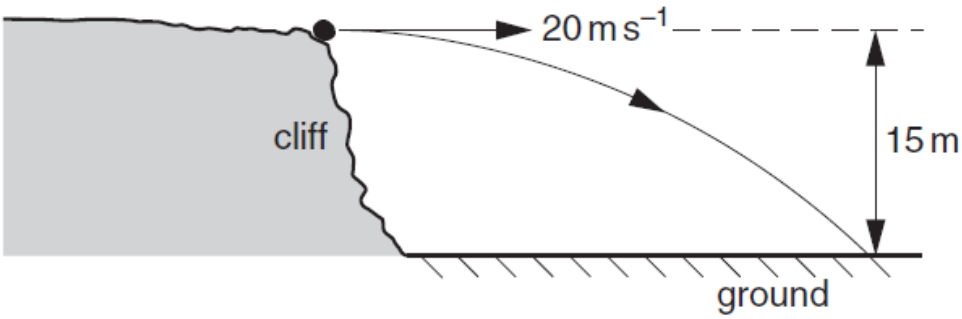

Fig. 1.1

Air resistance is negligible.

For this stone,

(i) calculate the time to fall 15 m,

time = ... s [2]

(ii) calculate the magnitude of the resultant velocity after falling 15 m,

resultant velocity = ... $m\,s^{-1}$ [3]